**5.3 EXERCISES**

**NAME : AIDIL ILHAM BIN AMINUDDIN**

**MATRIC NO : B032520007**

**1. Vending Machine Canned Drink & Coupon Calculator**

Assume we can buy a canned drink from the vending machine for RM1 each. Attached with every canned drink is a coupon. We can redeem 7 coupons for one canned drink from the machine. We would like to know how many canned drink can be redeemed, including those redeemed via coupon, if we have $n$ ringgit malaysia.

For example, if we have RM 20 then we can initially buy 20 canned drinks. This gives us 20 coupons. We can redeem 14 coupons for 2 additional canned drinks. These two additional canned drinks have 2 more coupons, so we now have a total of 8 coupons when added to the six leftover from the original purchase. This gives us enough to redeem for one final canned drink. As a result we now have 23 canned drinks and 2 leftover coupons.

Write a program that inputs the number of RM and outputs how many canned drink you can collect after spending all your money and redeeming as many coupons as possible. Also output the number of leftover coupons. The easiest way to solve this problem is to use a loop.

*Hint: Use a while loop that redeems coupons for canned drinks and calculates the new number of coupons. Continue the loop as long as you have enough coupons to redeem for a canned drink.*

Coding :

```
 #include <iostream>

using namespace std;

int main() {
    int n;
    cout << "Enter the amount of Ringgit Malaysia (RM): ";
    cin >> n;

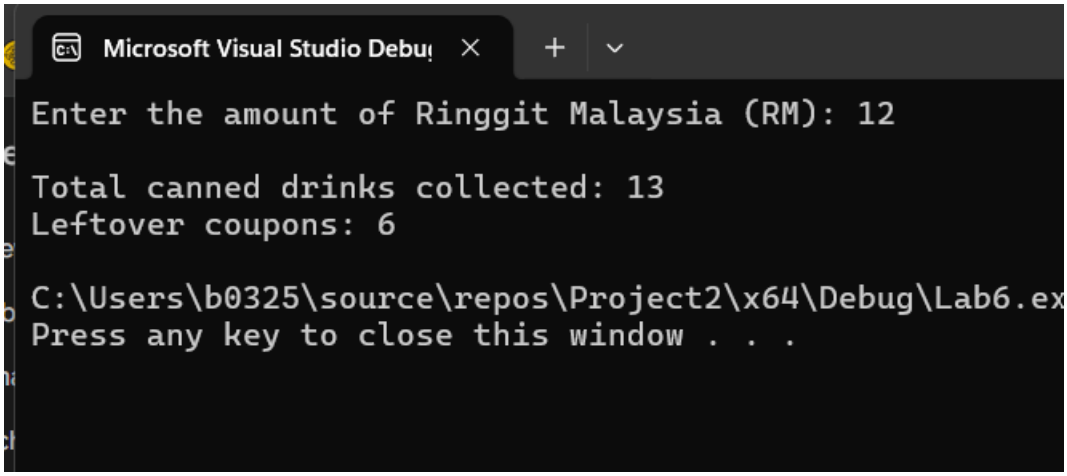
    int totalDrinks = n;
    int coupons = n;

    while (coupons >= 7) {
        int redeemedDrinks = coupons / 7; // Number of drinks we can get from coupons
        totalDrinks += redeemedDrinks;    // Add to total drinks collected
        coupons = (coupons % 7) + redeemedDrinks;
    }

    cout << "\nTotal canned drinks collected: " << totalDrinks << endl;
    cout << "Leftover coupons: " << coupons << endl;

    return 0;
}
```

Output :




**2. Sum of Integers from 1 to N**

Write a program that asks the user for a positive integer value. The program should use a loop to get the sum of all integers from 1 up to the number entered. For example, if the user enters 50, the loop will find the sum of 1,2,3,4,.... 50.

Coding :

```
#include <iostream>

using namespace std;

int main() {
    int n;
    cout << "Enter a positive integer: ";
    cin >> n;

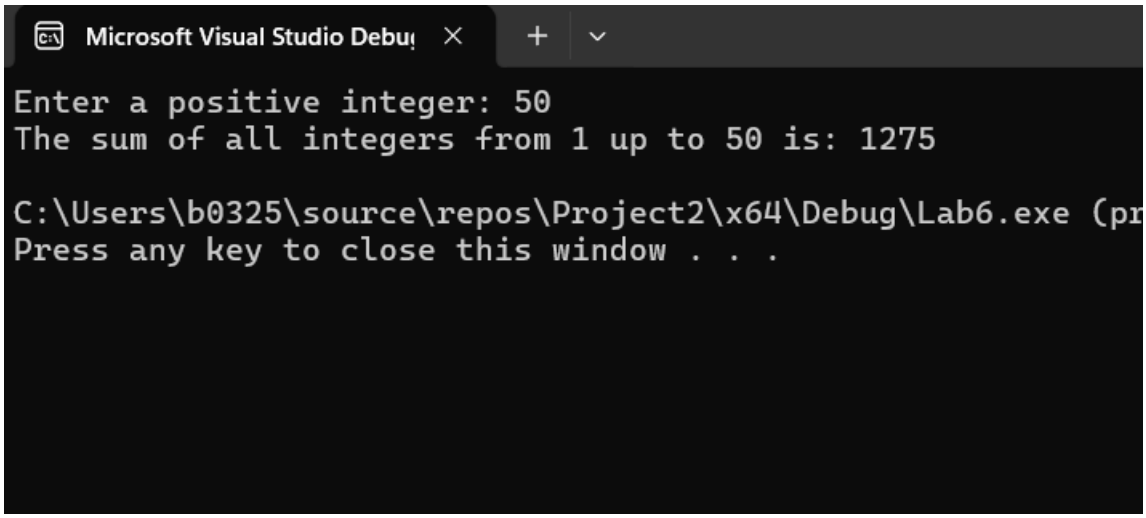
    // Validate for a positive integer
    while (n <= 0) {
        cout << "Please enter a positive integer: ";
        cin >> n;
    }

    int sum = 0;
    for (int i = 1; i <= n; ++i) {
        sum += i;
    }

    cout << "The sum of all integers from 1 up to " << n << " is: " << sum << endl;

    return 0;
}
```

Output :




**3. Monthly Budget Tracker**

Write a program that asks the user to enter the amount that he or she has budgeted for a month. The program should use a loop to prompt the user to enter each of his or her expenses for the month, and the program should be a able to keep the total running. When the loop finishes, the program should display whether the amount for the user is under or over the budget.

Coding :

```
#include <iostream>

using namespace std;

int main() {
    double budget;
    double expense;
    double totalExpenses = 0.0;

    cout << "Enter your monthly budget (RM): ";
    cin >> budget;

    cout << "\nEnter your monthly expenses one by one." << endl;
    cout << "Enter 0 or a negative number when you are finished." << endl;

    cout << "Enter expense: RM ";
    cin >> expense;

    while (expense > 0) {
        totalExpenses += expense;
        cout << "Enter next expense (or 0 to finish): RM ";
        cin >> expense;
    }

    cout << "\n--- Monthly Summary ---" << endl;
    cout << "Total Budget: RM " << budget << endl;
    cout << "Total Expenses: RM " << totalExpenses << endl;

    if (totalExpenses < budget) {
        cout << "You are under budget by RM " << (budget - totalExpenses) << "." <<
endl;
```

```
    }
    else if (totalExpenses > budget) {
        cout << "You are over budget by RM " << (totalExpenses - budget) << "!" <<
endl;
    }
    else {
        cout << "You have spent exactly your budget amount." << endl;
    }

    return 0;
}
```

Output :

